

Exploratory Data Analysis Report

Project: Healthcare Cost Estimator

Dataset Overview

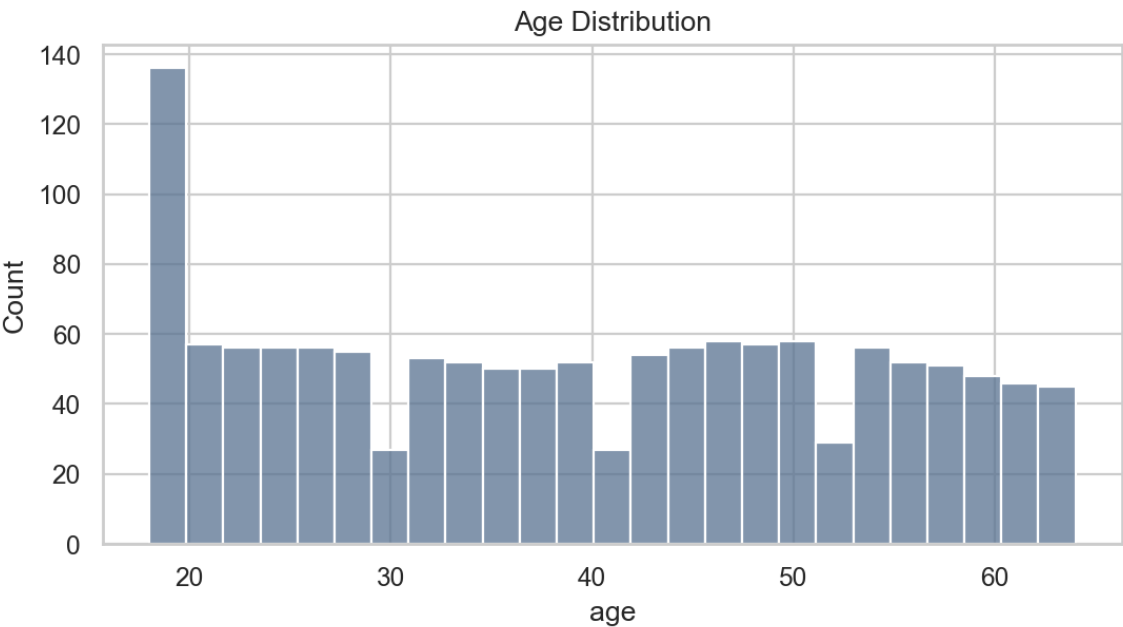
Original dataset records: 1338. Cleaned dataset records: 1337. Columns: age, sex, bmi, children, smoker, region, charges.

Dataset Statistics

index	age	bmi	children	charges
count	1337.0000	1337.0000	1337.0000	1337.0000
mean	39.2221	30.6635	1.0957	13279.1215
std	14.0443	6.1005	1.2056	12110.3597
min	18.0000	15.9600	0.0000	1121.8739
25%	27.0000	26.2900	0.0000	4746.3440
50%	39.0000	30.4000	1.0000	9386.1613
75%	51.0000	34.7000	2.0000	16657.7174
max	64.0000	53.1300	5.0000	63770.4280

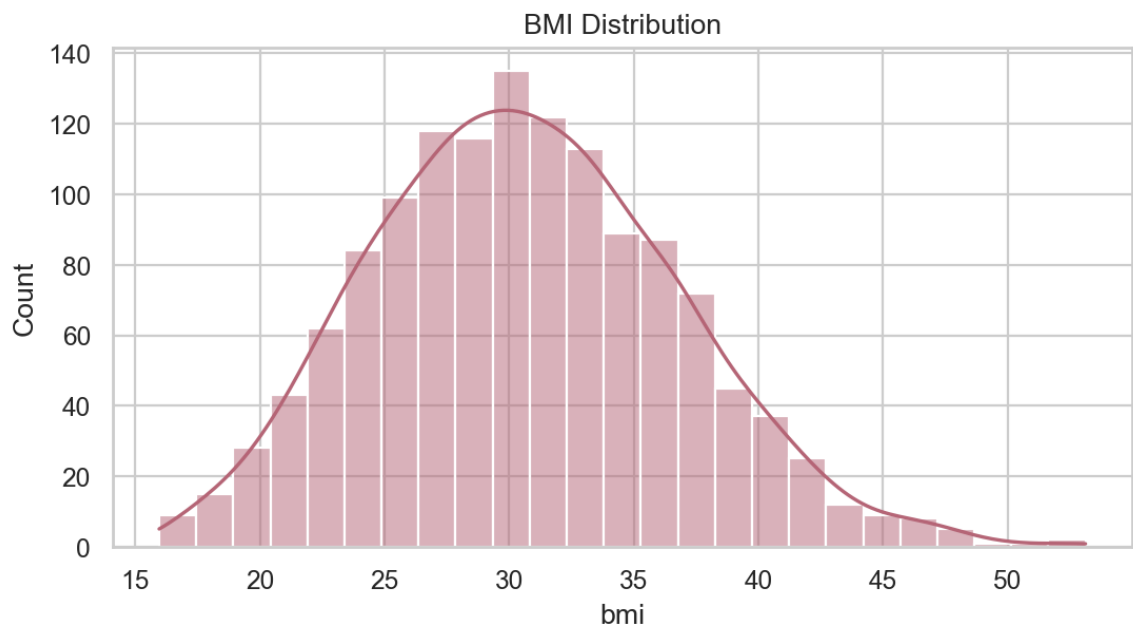
Age Analysis

Age ranges from 18 to 64 years, with a mean of 39.22.



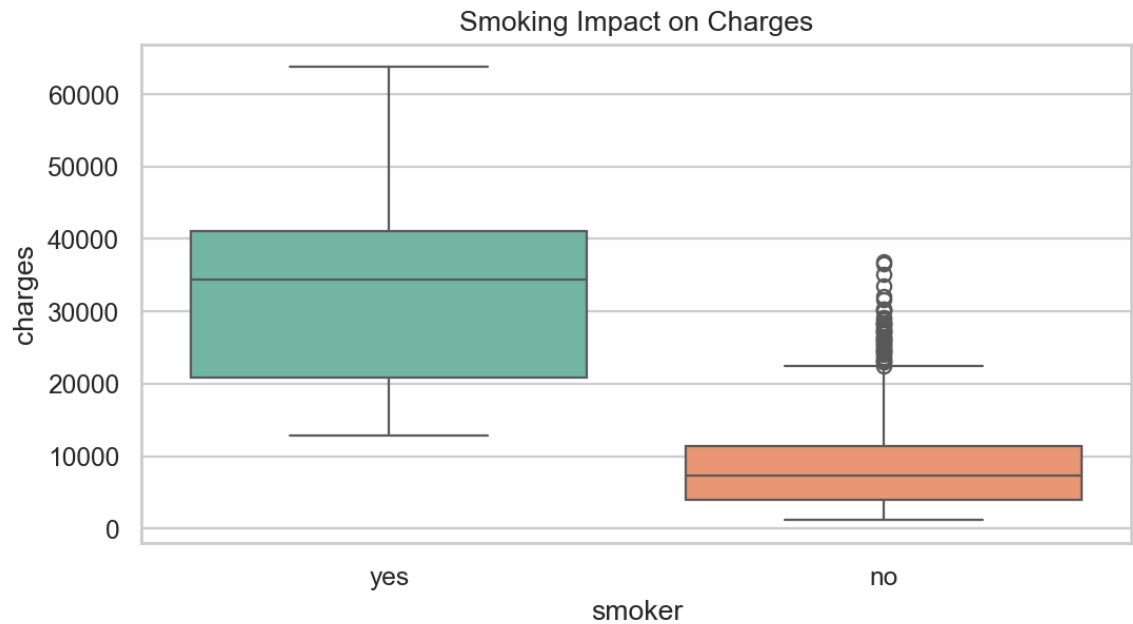
BMI Analysis

BMI ranges from 15.96 to 53.13, with a mean of 30.66.



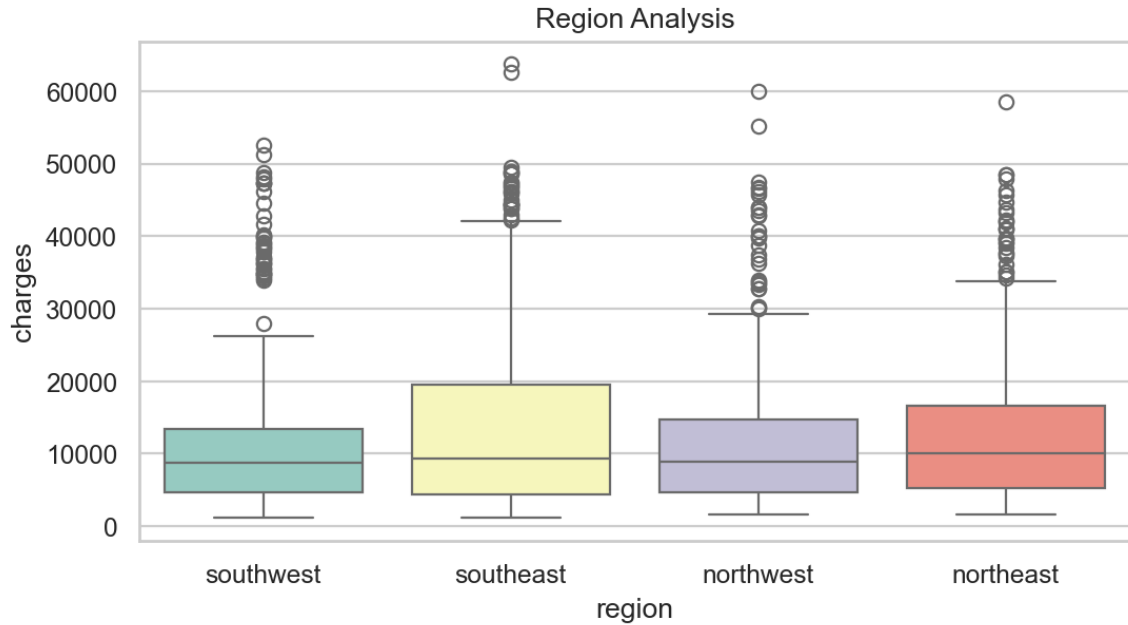
Smoking Impact Analysis

smoker	count	mean	median
no	1063	8440.6603	7345.7266
yes	274	32050.2318	34456.3485



Region Analysis

region	count	mean	median
northeast	324	13406.3845	10057.6520
northwest	324	12450.8408	8976.9772
southeast	364	14735.4114	9294.1319
southwest	325	12346.9374	8798.5930



Cost Distribution and Correlation

